Introduction to Generative AI

Overview of AI, ML, Deep Learning

What are LLMs and how they work

Applications of Generative AI in real world

Foundations Required

Python programming basics

Understanding APIs and JSON

Basic understanding of machine learning

Understanding LLMs

Tokens, embeddings, transformers

Popular models like GPT

Limitations of LLMs

Prompt Engineering

Zero-shot, few-shot prompting

Chain-of-thought reasoning

Best practices for prompts

Embeddings and Vector Databases

What are embeddings

Similarity search

Vector DBs like Chroma and FAISS

Retrieval Augmented Generation (RAG)

Why RAG is needed

Pipeline: retrieve + generate

Improving accuracy with RAG

LangChain Basics

Document loaders

Text splitters

Chains and retrievers

Building RAG Applications

Chat with PDF

Knowledge base assistants

Search systems

LangGraph for Agents

Stateful workflows

Graph-based reasoning

Agent decision making

LangSmith for Monitoring

Tracing and debugging

Observability

Evaluation of responses

Building Real-world AI Agents

Tool usage and APIs

Multi-agent systems

Autonomous workflows

Production & Scaling

Deployment strategies

Cost optimization

Scaling AI systems